

29 Beyond Perturbation Theory

29.3 Non-Perturbative Corrections to the Superpotential

We saw in Section 27.6 that the superpotential in general supersymmetric theories of gauge and chiral superfields is not renormalized to any finite order of perturbation theory, so that if supersymmetry is not broken in the tree approximation then it can only be broken by non-perturbative corrections to the Wilsonian effective Lagrangian. We now take up a general analysis of these corrections. These were thoroughly studied in a series of papers by Affleck, Davis, Dine, and Seiberg in the early 1980s [3], with special attention to the case of supersymmetric versions of quantum chromodynamics with arbitrary numbers of colors and flavors. Here we will present a somewhat simplified analysis of general supersymmetric gauge theories, based on the more recent holomorphy arguments of Seiberg [4] already used in Section 27.7.

To study non-perturbative effects, we shall here again consider a general renormalizable supersymmetric theory, but now including a possible θ -term in the Lagrangian density

$$\mathcal{L} = \left[\Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} + \operatorname{Re} \left[\frac{\tau}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \quad (29.3.1)$$

where the superpotential $f(\Phi)$ is a gauge-invariant cubic polynomial in the left-chiral superfields and τ is the parameter (27.3.23):

$$\tau = \frac{4\pi i}{g^2} + \frac{\theta}{2\pi}. \quad (29.3.2)$$

As in Section 27.6, we introduce a pair of gauge-invariant left-chiral external superfields, now called Y and T , and replace the Lagrangian density with

$$\mathcal{L}^\# = \left[\Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} [Y f(\Phi)]_{\mathcal{F}} + \operatorname{Re} \left[\frac{T}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}. \quad (29.3.3)$$

This becomes the same as (29.3.1) when we set the spinor and auxiliary components of Y and T equal to zero, and take their scalar components as $y = 1$ and $t = \tau$, respectively. Non-perturbative effects will in general invalidate both of the two symmetries on which the analysis of Section 27.6 was based. The translation operation, which in our present notation is $T \rightarrow T + \xi$ with real ξ , is not a symmetry because $\sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma}$ can have a non-vanishing integral over spacetime. The original R invariance (with T and Y having R values 0 and +2) is not a symmetry because the anomaly discussed in Chapter 22 gives the R -current a non-vanishing divergence. With θ_L and θ_R having $R = +1$ and $R = -1$, respectively, and V_A and Φ_n R -neutral, the fermion fields λ_{AL} and ψ_{nL} have $R = +1$ and $R = -1$, respectively, so Eq. (22.2.26) here gives

$$\partial_\mu J_R^\mu = -\frac{1}{32\pi^2} (C_1 - C_2) \sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma}, \quad (29.3.4)$$

where C_1 and C_2 are the constants defined in Eqs. (17.5.33) and (17.5.34):

$$\sum_{CD} C_{ACD} C_{BCD} = C_1 \delta_{AB}, \quad \text{Tr}\{t_A t_B\} = C_2 \delta_{AB}, \quad (29.3.5)$$

with the trace taken over all species of left-chiral superfield.¹ For instance, in the generalized supersymmetric version of quantum chromodynamics studied in Reference 3, with gauge group $SU(N_c)$ and N_f pairs of left-chiral quark superfields Q_n and $\bar{Q}_n$ in the defining representation and its complex conjugate, these constants have values given by Eq. (17.5.35) (with $n_f = 2N_f$) as

$$C_1 = N_c, \quad C_2 = N_f.$$

Although T translation and R invariance are invalidated by non-perturbative effects, there is a remaining symmetry which is almost as powerful. Consider a general R transformation

$$\theta_L \longrightarrow e^{i\varphi} \theta_L, \quad \Phi \longrightarrow \Phi, \quad V_A \longrightarrow V_A, \quad Y \longrightarrow e^{2i\varphi} Y, \quad (29.3.6)$$

with arbitrary real φ . This leaves the T -independent terms of the Lagrangian density (29.3.3) invariant, but according to Eq. (29.3.4), quantum effects violate this symmetry, just as if there were a term $\Delta\mathcal{L}$ in the Lagrangian density with a transformation

$$\Delta\mathcal{L} \longrightarrow \Delta\mathcal{L} - \frac{1}{32\pi^2} (C_1 - C_2) \sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma} \varphi.$$

Recalling Eq. (27.3.18), this is cancelled if we give T a transformation

$$T \longrightarrow T + (C_1 - C_2) \varphi / \pi. \quad (29.3.7)$$

Because $W_{A\alpha L}$ has $R = 2$, the whole theory including non-perturbative effects is invariant under the combined transformations (29.3.6) and (29.3.7). In particular, the superfield $\exp(2i\pi T)$, which for $T = \tau$ is periodic in θ , has $R = 2(C_1 - C_2)$.

We again introduce an ultraviolet cut-off, and consider the effective ‘Wilsonian’ Lagrangian

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, T, T^\dagger, Y, Y^\dagger, \mathcal{D} \cdots) \right]_D \\ & + 2 \text{Re} \left[\frac{T}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} + \mathcal{B}_\lambda(\Phi, W_L, T, Y) \right]_{\mathcal{F}}, \end{aligned} \quad (29.3.8)$$

with $\mathcal{A}_\lambda$ and $\mathcal{B}_\lambda$ both gauge-invariant functions of the displayed arguments. The term proportional to T has been separated from the function $\mathcal{B}_\lambda$ in order that the translation (29.3.7) of T should continue to cancel the anomaly in the R transformation (29.3.6).

¹A factor 32 instead of 16 appears in the denominator in Eq. (29.3.4) because gauginos do not have distinct antiparticles, and we are now counting antiparticles separately from particles in taking the trace in Eq. (29.3.5). Also, we are now adopting the convention, described at the end of Section 27.3, of including a gauge coupling factor in the gauge fields and not in structure constants and the matrix generators t_A . The gauge generators are thus normalized so that for t_A , t_B , and t_C in the standard $SU(2)$ subalgebra of the gauge algebra, the structure constant is $C_{ABC} = \epsilon_{ABC}$.

Invariance under the combined transformation (29.3.6), (29.3.7) then tells us that terms in the function $\mathcal{B}_\lambda$ must be proportional to powers of $\exp(2i\pi T)$, which have definite R values.

Furthermore, it is only *positive* powers of $\exp(2i\pi T)$ that may appear in $\mathcal{B}_\lambda$. According to Eq. (27.3.24), it is only instantons with positive winding number $\nu \geq 0$ that can make contributions to the effective Lagrangian that are holomorphic in T rather than T^* , and these give rise to factors $\exp(2i\pi\nu T)$. More generally, for $T = \tau$ any power $\exp(2ia\pi T)$ will depend on the gauge coupling through a factor $\exp(-8\pi^2 a/g^2)$, so that a must be positive in order for non-perturbative effects to be suppressed for small g . In consequence, non-perturbative effects enter in $\mathcal{L}_\lambda^\#$ through operators $\exp(2i\pi a T)$ that have positive-definite, zero, or negative-definite values of R , depending on whether $C_1 > C_2$, $C_1 = C_2$, or $C_1 < C_2$. (In the generalized supersymmetric version of quantum chromodynamics described above, this corresponds to $N_c > N_f$, $N_c = N_f$, and $N_c < N_f$, respectively.) We shall now consider each of these cases.

$$C_1 > C_2$$

Here the powers $\exp(2ia\pi T)$ with $a > 0$ have positive-definite values of R , given by Eq. (29.3.7) as $R = 2(C_1 - C_2)a$. Lorentz invariance tells us that if any term in $\mathcal{B}_\lambda$ contains a factor $W_{A\alpha L}$, then it must contain at least two of them, so the only ways to construct terms in $\mathcal{B}_\lambda$ with $R = 2$ is to have two W s and no dependence on Y or T , or one Y and no dependence on W or T , or one factor of $\exp(2i\pi T/(C_1 - C_2))$ and no dependence on W or Y :

$$\begin{aligned} \mathcal{B}_\lambda = & Y f_\lambda(\Phi) + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta B} \ell_{\lambda AB}(\Phi) \\ & + \exp\left(\frac{2i\pi T}{C_1 - C_2}\right) v_\lambda(\Phi). \end{aligned} \quad (29.3.9)$$

Because $f_\lambda(\Phi)$ does not depend on Y or T , it can only be the tree-approximation superpotential

$$f_\lambda(\Phi) = f(\Phi), \quad (29.3.10)$$

just as in perturbation theory. Likewise, because $\ell_{\lambda AB}(\Phi)$ does not depend on Y or T it must have equal numbers of Φ s and $\Phi^\dagger$ s, so since it does not depend on $\Phi^\dagger$ it cannot depend on Φ either. Gauge invariance then requires (for a simple gauge group) that $\ell_{\lambda AB}(\Phi)$ is proportional to δ_{AB} , and since it does not depend on T or Y , the power-counting argument of Section 27.6 shows that the coefficient of δ_{AB} can only be the one-loop contribution to the running inverse-square Wilsonian gauge coupling.

To be more explicit about the running gauge coupling, recall that in non-supersymmetric gauge theories with fermions, Eq. (18.7.2) gives the one-loop renormalization group equation as

$$\lambda \frac{dg_\lambda}{d\lambda} = b g_\lambda^3, \quad (29.3.11)$$

with

$$b = -\frac{1}{4\pi^2} \left(\frac{11}{12} C_1 - \frac{1}{6} C_2 \right), \quad (29.3.12)$$

with the coefficient of C_2 taken as $-1/6$ rather than $-1/3$ because we are now counting the left-chiral states of antifermions separately from the left-chiral states of particles. As we have seen in Section 28.2, the effect of gauginos is to multiply the C_1 -term by a factor $9/11$, while the effect of the scalar components of the left-chiral superfields (such as squarks and sleptons) is to multiply the C_2 -term by a factor $3/2$, so in supersymmetric theories Eq. (29.3.12) becomes instead

$$b = -\frac{1}{16\pi^2}(3C_1 - C_2). \quad (29.3.13)$$

The solution of Eq. (29.3.11) for the running gauge coupling is then

$$g_\lambda^{-2} = g^{-2} + \frac{3C_1 - C_2}{8\pi^2} \ln \left(\frac{\lambda}{K} \right), \quad (29.3.14)$$

where K is an ultraviolet cut-off, introduced to give meaning to the otherwise ultraviolet-divergent bare gauge coupling g .

To summarize the results so far, setting $T = \tau$ and $Y = 1$, the Wilsonian effective Lagrangian for $C_1 > C_2$ takes the form

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, \tau, \tau^*, \mathcal{D} \dots) \right]_D + 2 \operatorname{Re} \left[\frac{\tau_\lambda}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}} \\ & + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} + \exp \left(\frac{2i\pi\tau_\lambda}{C_1 - C_2} \right) [v_\lambda(\Phi)]_{\mathcal{F}}, \end{aligned} \quad (29.3.15)$$

where

$$\tau_\lambda = \frac{4\pi i}{g_\lambda^2} + \frac{\theta}{2\pi}. \quad (29.3.16)$$

We have been able to replace τ with τ_λ in the exponential in the last term in Eq. (29.3.15), because the difference is a constant times $\ln \lambda$, which yields a power of λ that can be absorbed into the definition of v_λ .

Non-perturbative effects have now been isolated in the last term in Eq. (29.3.15). This term can be generated by instantons of winding number $\nu > 0$ if $C_1 - C_2 = 1/\nu$. (In general $C_1 - C_2$ is a rational number. For the generalized supersymmetric version of quantum chromodynamics, $C_1 - C_2 = N_c - N_f$ is an integer, so the condition $C_1 - C_2 = 1/\nu$ requires that $N_c = N_f - 1$, and then only $\nu = 1$ instantons contribute. Detailed calculations [5] in this model show that instantons actually do make such contributions.) Whether or not it is instantons that generate the non-perturbative contribution $v_\lambda(\Phi)$, we can determine its form by considering the non-anomalous symmetries of the theory. Since this function is independent of Y , it can be evaluated as if $Y = 0$, so it shares all the non-anomalous symmetries of the first term in Eq. (29.3.1). These include the gauge symmetry itself and a global symmetry under $\prod_d SU(n(d))$, where d labels the different irreducible representations of the gauge group furnished by the left-chiral superfields, and $n(d)$ is the number of times representation d occurs. (For instance, in generalized supersymmetric quantum chromodynamics d takes two values, labelling the N_c and $\bar{N}_c$ representations of $SU(N_c)$,

and $n(N_c) = n(\bar{N}_c) = N_f$.) Let us label the Φ s as $\Phi_{ai}^{(d)}$, where a is a gauge index, and i is a ‘flavor’ index, labelling the $n(d)$ different Φ s that transform under the gauge group according to the representation d . The only way to construct a function of the Φ s that is invariant under the global symmetry group $\prod_d SU(n(d))$ is as a product of Φ s, with the $n(d)$ flavor indices contracted for each d with the antisymmetric $SU(n(d))$ tensor $\epsilon_{i_1 \dots i_{n(d)}}$, and with the gauge indices contracted with constant tensors of the gauge group. (For instance, for generalized supersymmetric quantum chromodynamics, v_λ must be a function of the sole invariant

$$D \equiv \text{Det}_{ij} \sum_a Q_{ai} \bar{Q}_{aj} ,$$

which is non-zero only for $N_c \geq N_f$.)

In addition to the anomaly-free $SU(n(d))$ flavor symmetries, there is also a $U_d(1)$ symmetry for each of the irreducible representations d furnished by the Φ s, with all $\Phi_{ai}^{(d)}$ for a given d undergoing the transformation

$$\Phi_{ai}^{(d)} \longrightarrow e^{i\varphi_d} \Phi_{ai}^{(d)} . \quad (29.3.17)$$

This symmetry is anomalous, with the effects of the anomaly the same as if the Lagrangian underwent the change

$$\mathcal{L} \longrightarrow \mathcal{L} - \sum_d \frac{n(d)C_{2d}}{32\pi^2} \sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma} \varphi_d , \quad (29.3.18)$$

where C_{2d} is the contribution to C_2 of any one left-chiral scalar superfield belonging to the irreducible representation d of the gauge group. The symmetry is restored if we give T the transformation property

$$T \longrightarrow T + n(d)C_{2d}\varphi_d/\pi . \quad (29.3.19)$$

Since $v_\lambda(\Phi)$ is accompanied in Eq. (29.3.9) by a factor $\exp(2i\pi T/(C_1 - C_2))$, which undergoes the transformation

$$\exp\left(\frac{2i\pi T}{C_1 - C_2}\right) \longrightarrow \prod_d \exp\left(\frac{+2in(d)C_{2d}\varphi_d}{C_1 - C_2}\right) \exp\left(\frac{2i\pi T}{C_1 - C_2}\right) , \quad (29.3.20)$$

we conclude that *for each representation d of the gauge group furnished by the left-chiral scalars, $v_\lambda(\Phi)$ must be a homogeneous function of the $\Phi_{ai}^{(d)}$ of negative order $-2n(d)C_{2d}/(C_1 - C_2)$* . (For instance, in generalized supersymmetric quantum chromodynamics we have two irreducible representations of $SU(N_c)$, the defining and antidefining representations, each with $n(d) = N_f$ and $C_{2d} = 1/2$, so v_λ is a homogeneous function of order $-N_f/(N_c - N_f)$ in the Q belonging to the defining representation and of the same order in the $\bar{Q}$ belonging to the antidefining representation. Thus it must be proportional to $D^{-1/(N_c - N_f)}$, where D is the determinant introduced earlier.) In general $C_2 = \sum_d n(d)C_{2d}$, so $v_\lambda(\Phi)$ is a homogeneous function of all the Φ s, of order $-2C_2/(C_1 - C_2)$.

This result satisfies an important consistency check. Recall from Section 27.4 that any superpotential has dimensionality $+3$ (counting powers of mass, with $\hbar = c = 1$), while the

scalar superfields Φ like ordinary scalar fields have dimensionality $+1$, so the Φ -dependent part of v_λ must appear with a coefficient of dimensionality

$$3 + \frac{2C_2}{C_1 - C_2} = \frac{3C_1 - C_2}{C_1 - C_2}.$$

This coefficient does not depend on the gauge coupling or on any of the couplings or masses in the superpotential, and because we have replaced the bare coupling g with g_λ in the second term of Eq. (29.3.11) it cannot depend on the ultraviolet cut-off K used to define g either, so it can only depend on λ . Therefore

$$v_\lambda(\Phi) = \lambda^{(3C_1 - C_2)/(C_1 - C_2)} H(\Phi), \quad (29.3.21)$$

where $H(\Phi)$ is a homogeneous function of order $-2n(d)C_{2d}/(C_1 - C_2)$ in the Φ s that belong to each representation d of the gauge group and is independent of any parameters of the theory. We can rewrite Eq. (29.3.14) in the form

$$\tau_\lambda = i \frac{3C_1 - C_2}{2\pi} \ln \left(\frac{\lambda}{\Lambda} \right) + \frac{\theta}{2\pi}, \quad (29.3.22)$$

where Λ is an energy parameter that characterizes the running gauge coupling, like the $\Lambda \simeq 200$ MeV of quantum chromodynamics. Thus the last term in Eq. (29.3.15) is

$$\exp \left(\frac{2i\pi\tau_\lambda}{C_1 - C_2} \right) v_\lambda(\Phi) = \exp \left(\frac{i\theta}{C_1 - C_2} \right) \Lambda^{(3C_1 - C_2)/(C_1 - C_2)} H(\Phi). \quad (29.3.23)$$

The whole effective superpotential, including the non-perturbative contribution (29.3.23), is therefore independent of the floating cut-off λ .

The function $H(\Phi)$ is homogeneous and of negative order in Φ , so in the absence of a bare superpotential the potential is positive-definite at finite values of the scalar fields and vanishes only at infinite field values. In such a theory, there is no stable vacuum state, and the question of supersymmetry breaking is moot. The vacuum may be stabilized by adding a suitable bare superpotential. For instance, in generalized supersymmetric quantum chromodynamics with $N_f < N_c$ the only renormalizable bare superpotential is a sum of mass terms

$$f(Q, \bar{Q}) = \sum_{ija} m_{ij} \bar{Q}_{ai} Q_{aj}. \quad (29.3.24)$$

To seek a supersymmetric vacuum state, we need first of all to find what scalar components q_{ai} and $\bar{q}_{ai}$ satisfy Eq. (27.4.11), which here reads

$$\sum_{abi} q_{ai}^* (t_A)_{ab} q_{bi} - \sum_{abi} \bar{q}_{ai}^* (t_A)_{ab} \bar{q}_{bi} = 0, \quad (29.3.25)$$

for all generators t_A of $SU(N_c)$. The gauge interactions (but not the superpotential) are invariant under simultaneous $SU(N_c)$ transformations on the color indices a of both q_{ai} and $\bar{q}_{ai}$; under independent $SU(N_f)$ and $\overline{SU(N_f)}$ transformations on the flavor indices i of q_{ai} and $\bar{q}_{ai}$, respectively, and under a $U(1)$ transformation of both q_{ai} and $\bar{q}_{ai}$ by opposite

phases. Using these symmetries, it is possible to put the general solution of these conditions in the form

$$q_{ai} = \bar{q}_{ai} = \begin{cases} u_i \delta_{ai}, & a \leq N_f, \\ 0, & a > N_f, \end{cases} \quad (29.3.26)$$

where the u_i are complex numbers with the same phase. (Here is the proof. The $SU(N_c)$ generators t_A span the space of all traceless Hermitian matrices, so Eq. (29.3.25) is equivalent to the requirement that

$$\sum_i q_{ai}^* q_{bi} - \sum_i \bar{q}_{ai}^* \bar{q}_{bi} = k \delta_{ab}, \quad (29.3.27)$$

for some constant k . By a combined color and flavor transformation $q \rightarrow UqV$ with U and V unitary and unimodular, we can put the matrix q in the diagonal form (29.3.26), and by a unimodular change of phase of the diagonal elements we can arrange that they all have the same phase. Then Eq. (29.3.27) becomes

$$\sum_i \bar{q}_{ai}^* \bar{q}_{bi} = \begin{cases} (u_a^2 - k) \delta_{ab}, & a \leq N_f, \\ -k \delta_{ab}, & a > N_f. \end{cases}$$

The conditions for $a > N_f$ show that $k \leq 0$. If k were non-zero, then the $\bar{q}_{ai}$ would furnish N_c non-zero orthogonal vectors with N_f components, which is impossible for $N_f < N_c$, so $k = 0$. We can then put the $\bar{q}_{ai}$ in the diagonal form (29.3.26) by a series of unitary flavor transformations: first rotate $\bar{q}_{1i}$ into the 1-direction; then keeping the 1-direction fixed, rotate in the space perpendicular to this direction to put $\bar{q}_{2i}$ in the 2-direction; and so on; and then perform a unimodular phase transformation so that all diagonal elements have the same phase. Eq. (29.3.27) then shows that the absolute values of the diagonal elements of q_{ai} and $\bar{q}_{ai}$ are equal, and by a non-anomalous opposite phase change of q_{ai} and $\bar{q}_{ai}$ we can arrange that their common phases are equal, as was to be proved.)

The function H in Eq. (29.3.23) is here

$$H(q, \bar{q}) = \mathcal{J} \left[\text{Det}_{ij} \sum_a q_{ai} \bar{q}_{aj} \right]^{-1/(N_c - N_f)} = \mathcal{J} \left[\prod_i u_i \right]^{-2/(N_c - N_f)}, \quad (29.3.28)$$

where $\mathcal{J}$ is a purely numerical constant. (Detailed calculations show that $\mathcal{J} = 2$ for $N_f = 2$ and $N_c = 3$.) Adding the terms (29.3.23) and (29.3.24), the complete effective superpotential is now

$$f_{\text{total}}(q, \bar{q}) = \mathcal{K} \left[\prod_i u_i \right]^{-2/(N_c - N_f)} + \sum_i m_i u_i^2, \quad (29.3.29)$$

where

$$\mathcal{K} \equiv \mathcal{J} \exp \left(\frac{i\theta}{N_c - N_f} \right) \Lambda^{(3N_c - N_f)/(N_c - N_f)} \quad (29.3.30)$$

and the m_i are the diagonal elements of the mass matrix that results when the original mass matrix is subjected to the $SU(N_f) \cdot \overline{SU(N_f)}$ transformation used to put the scalars in

the form (29.3.26). The condition (27.4.10), that $f_{\text{total}}(q, \bar{q})$ should be stationary, has the solution

$$u_i^2 = \frac{1}{m_i} \left(\frac{\mathcal{K}}{N_c - N_f} \right)^{1 - N_f/N_c} \left(\prod_j m_j \right)^{-1/N_c}. \quad (29.3.31)$$

Because we have put the scalar fields in a basis in which the u_i have a common phase, the m_i must also have a common phase in this basis. But the common phase of the u_i^2 is not unique—the $1/N_c$ powers in Eq. (29.3.31) tell us that the solution is undetermined by a factor $\exp(2i\pi n/N_c)$, with n an integer ranging from 0 to $N_c - 1$. (The two signs of u_i for a given u_i^2 are physically equivalent, because the whole theory is invariant under a non-anomalous symmetry with $q_{ai} \rightarrow e^{i\pi} q_{ai}$ and $\bar{q}_{ai} \rightarrow e^{-i\pi} \bar{q}_{ai}$.) The fact that there are N_c physically inequivalent solutions will turn up again in our discussion of the Witten index in the next section.

$$C_1 = C_2$$

This case is of some interest because, as we saw in Eq. (27.9.3), the simplest $N = 2$ supersymmetric Yang–Mills theory, when written in terms of $N = 1$ superfields, contains a single left-chiral superfield in the adjoint representation, for which² of course $C_2 = C_1$.

For $C_2 = C_1$ the function $\exp(2i\pi T)$ has $R = 0$, so that its appearance in $\mathcal{L}_\lambda^\#$ is not restricted by R invariance. The general form of the $\mathcal{F}$ -term in $\mathcal{L}_\lambda^\#$ is given here by Eq. (29.3.9), but with the last term absent:

$$\mathcal{B}_\lambda = Y f_\lambda(\Phi, \exp(2i\pi T)) + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta B} \ell_{\lambda AB}(\Phi, \exp(2i\pi T)). \quad (29.3.32)$$

Because f_λ may depend on T , we cannot now conclude that it is equal to the bare superpotential, but only that it depends linearly on whatever coupling coefficients and masses appear linearly in the bare superpotential. In particular, *if there is no superpotential to begin with, then none is generated by non-perturbative effects.*

To go further, we need to make use of the anomalous chiral symmetry under a $U(1)$ transformation of all the Φ_n . In order for the whole theory to be invariant under this symmetry, it is necessary to introduce a separate external left-chiral superfield Y_r for the terms in the bare superpotential of order r in the Φ_n . Then the theory is invariant under the combined transformations

$$\Phi_n \rightarrow e^{i\varphi} \Phi_n, \quad T \rightarrow T + C_2 \varphi / \pi, \quad Y_r \rightarrow e^{-ir\varphi} Y_r. \quad (29.3.33)$$

This symmetry tells us that a term in $\mathcal{B}_\lambda$ that is of order $\mathcal{N}_r$ in the coefficients of the term in the superpotential of order r in Φ and proportional to $\exp(2ai\pi T)$ must be of an order $\mathcal{N}_\Phi$ in Φ , given by

$$\mathcal{N}_\Phi = \sum_r r \mathcal{N}_r - 2C_2 a. \quad (29.3.34)$$

²Note that here C_2 refers to the representation of the gauge group furnished by the chiral superfields, so it is the same as the quantity C_2^b in Section 27.9, which refers to the representation furnished by complex scalars, but half of C_2^f , which refers to the representation furnished by all spinor fields, including the gauginos.

The coefficients $\ell_{\lambda AB}$ of the terms in $\mathcal{B}_\lambda$ that are quadratic in W are shown in Eq. (29.3.32) to be independent of the parameters in the superpotential for $C_1 = C_2$, so in this case Eq. (29.3.34) becomes

$$\mathcal{N}_\Phi = -2C_2 a. \quad (29.3.35)$$

Thus there can be no terms in $\ell_{\lambda AB}$ of positive order in the Φ_r , and any term in $\ell_{\lambda AB}$ that is independent of the Φ_r must be independent of T . These Φ -independent terms in $\ell_{\lambda AB}$ are therefore again just the one-loop contribution to the running coupling parameter τ_λ .

The effective superpotential is shown by Eq. (29.3.32) to be linear in the parameters in the superpotential, so all of its terms have just one $\mathcal{N}_r = 1$ and the others zero. For such a term Eq. (29.3.34) gives

$$\mathcal{N}_\Phi = r - 2C_2 a. \quad (29.3.36)$$

A term in the effective superpotential with $\mathcal{N}_\Phi$ powers of Φ can therefore only arise from terms in the bare superpotential with $r \geq \mathcal{N}_\Phi$ powers of Φ . The terms with $r = \mathcal{N}_\Phi$ have $a = 0$, so they are given by the tree approximation as just the bare superpotential. The only other terms are non-perturbative corrections with $r > \mathcal{N}_\Phi$. Such a non-perturbative term, of a given order in the Φ s, can only arise from terms in the bare superpotential of *higher* order in the Φ s.

$$C_1 < C_2$$

Here the R value $2(C_1 - C_2)$ of $\exp(2i\pi T)$ is *negative*, so positive powers of $\exp(2i\pi T)$ can compensate for the positive R values of Y and W_α , and $\mathcal{B}_\lambda$ may therefore contain terms of arbitrary order in Y and W_α . Using the chiral symmetry condition (29.3.34) and the R invariance condition

$$2 = \mathcal{N}_W + 2 \sum_r \mathcal{N}_r - 2a(C_2 - C_1), \quad (29.3.37)$$

we can, however, set limits on the structure of terms of a given order $\mathcal{N}_\Phi$ in the Φ s. Eqs. (29.3.34) and (29.3.37) have a trivial solution with $\mathcal{N}_r = 1$ for $r = \mathcal{N}_\Phi$, $\mathcal{N}_r = 0$ for other values of r , and $a = \mathcal{N}_W = 0$; this solution just represents the presence in $\mathcal{B}_\lambda$ of the original bare superpotential, with no radiative corrections. If there is no superpotential to begin with, then $\mathcal{N}_r = 0$ for all r , so Eq. (29.3.35) does not allow any terms in the Wilsonian Lagrangian with $\mathcal{N}_W = 0$, and so *no superpotential can be generated*. (In the supersymmetric version of quantum chromodynamics this conclusion is usually derived by noting that there is no possible term in the superpotential that would be consistent with all symmetries, but as we have now seen, the conclusion is much more general.) For renormalizable asymptotically free theories there is a useful limit on the structure of Φ -independent terms in $\mathcal{B}_\lambda$. The condition of renormalizability tells us that $\mathcal{N}_r = 0$ for $r > 3$, so by subtracting 2/3 of Eq. (29.3.34) from Eq. (29.3.37) we find

$$2 \geq \frac{2}{3} \mathcal{N}_\Phi + \mathcal{N}_W + 2a \left(C_1 - \frac{1}{3} C_2 \right). \quad (29.3.38)$$

Asymptotic freedom requires that $3C_1 > C_2$, so for $\mathcal{N}_\Phi = 0$ (or $\mathcal{N}_\Phi > 0$) each term on the right-hand side is positive. It follows that we can have no Φ -independent terms of higher

than second order in W , and these terms are also independent of T , so they again represent the one-loop contribution to the running coupling parameter τ_λ . But for $C_1 < C_2$ there is no general prohibition against terms of second or higher order in W and negative order in Φ .